

Installation Survey

Chassis Model	Sales designation	Brake system Trailer ¹⁾	Installation
435.115	U 1300 L	Dual-line	SA 36 043
435.117	U 1300 L/37	Single- and dual-line	

1) For brake system tractor unit, see 42.11-1.1/1.

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Notes on Plastic Pipelines**Special Tools**

Cons. No.	Tool	Number	Tool Set
1	Clamp	387 589 02 37 00	B

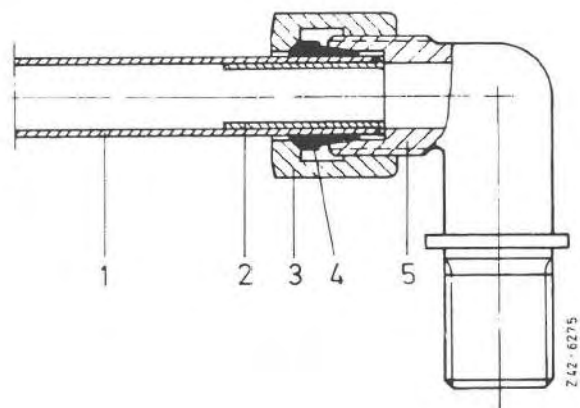
Material:

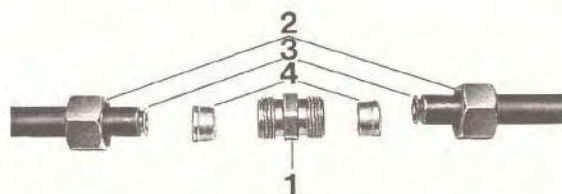
Plastic pipelines are made of polyamide 11 or 12 (W = soft, LT = light and thermally stable). Polyamide is flexible and resists all fluids, oils and greases used in motor vehicles. In addition, plastic piping is also resistant to alkalines, unchlorinated solvents, organic and inorganic acids and diluted oxidents. The melting point of plastic piping is $+180^{\circ}\text{C}$. Normal operating temperatures lie between -40°C and $+100^{\circ}\text{C}$.

Plastic piping should be protected or removed prior to drilling and welding work.

Coupling and Fitting:

When preparing every new screw joint, the plastic pipe (1) must be cut at right angles. The sleeve (2) is then inserted as far as it will go in the end of the pipe, the union nut (3) and the bite ring (4) are then slid on to the plastic pipe. Now hand-tighten union nut until appreciable resistance is felt at pipe socket (5). Using wrench, tighten by a further $1\frac{1}{2}$ to $1\frac{3}{4}$ turns. Release union nut and check whether bite ring is seated firmly in plastic pipe and also whether a slight bulge has appeared in front of the bite ring. Retighten union nut in the normal manner.





Damage to plastic piping does not imply replacement of the entire line. Damaged sections can be cut out and repaired with the aid of a double adapter.

R47-6001

- 1 Double adapter
- 2 Nut
- 3 Sieve insert
- 4 Bite ring

Care must be taken when disassembling and assembling **inlet union impact connections** to ensure that the sealing edges of the sawtooth-shaped retaining ribs are not damaged otherwise leaks may occur. Damaged connections must be renewed.

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RU 6000-12

- 1 Hold plastic pipe in clamp.

- 2 Place inlet union connection on end of pipe and drive in as far as it will go with plastic-headed hammer.

Note:

The end of the pipe **must not** be heated up (room temperature).

The tip of the inlet should be lightly oiled in the case of thin-walled or pipes of a small diameter thereby facilitating the entry of the inlet union.

Troubleshooting

Compressed air system (EEC system)

Repairs to assemblies may only be executed by authorized workshops and brake service points.

Fault	Cause	Remedy
Air compressor/auxiliary air compressor		
Compressed air delivery inadequate or non-existent.	Suction or delivery valves leaking. Excessive play between piston and cylinder. Line or filter in pressure regulator coked up. Cylinder head gasket defective.	Overhaul or exchange air compressor. Clean line or filter in pressure regulator, exchanging if necessary. Exchange cylinder head gasket.
Air compressor overheating.	Brake system leaking heavily.	Rectify leakage in brake system.
	Line or filter from pressure regulator coked up.	Clean line or filter from pressure regulator, exchanging if necessary.
	Too many additional air loads.	Install auxiliary air compressor.
Air compressor using too much oil	Vacuum in intake line.	Clean air filter, exchanging filter element if necessary.
	Cylinder and piston rings heavily worn.	Overhaul or exchange air compressor.
Pressure regulator		
At charging position, compressed air escapes at outlet socket.	No load valve leaking.	Clean or exchange valve and valve seat.
Pressure regulator fails to switch air compressor over to no-load.	Diaphragm piston sticking.	Check or exchange diaphragm piston.
Pressure regulator fails to switch air compressor over to on-load.	Adjusting spring defective, cut-off piston sticking.	Exchange spring, check cut-off piston and sealing ring.
At charging position, pressure regulator loses compressed air at vent.	Intake valve leaking, spring defective.	Clean or exchange valve, exchange spring.
At no-load position, pressure regulator loses compressed air at vent.	Exhaust valve leaking, spring defective.	Clean or exchange valve, exchange spring.
Pressure regulator does not admit compressed air to air tank.	Filter choked.	Clean or exchange filter.
Brief switching interval between pressure regulator cut-in and cut-out.	High air consumption, check valve leaking, sealing ring or exhaust valve leaking.	Check brake system for leakage. Check valves and seals. Exchange wear parts.

Fault	Cause	Remedy
Defroster		
Compressed air escapes at actuating rod.	Sealing ring leaking.	Exchange sealing ring at actuating rod.
At closed position, steady consumption of alcohol.	Sealing ring leaking.	Exchange sealing ring.
Defroster does not work satisfactorily at operating position.	No antifreeze left. More water than antifreeze in supply tank.	Drain water and fill up with antifreeze.
Four-circuit protection valve		
Compressed air escapes at breather hole.	Seal in valve body leaking.	Check or replace seal.
Opening pressure of individual valve circuits not within tolerance range.	Incorrect adjustment	Check valve setting on test bench. Replace spring and set on test bench.
	Spring defective.	
	Valve body sticking or jammed.	Exchange 4-circuit protection valve.
With failure in first or second circuit, pressure not maintained in third or fourth circuit.	Check valves leaking or defective.	Check or exchange check valves.
Drain valve		
Valve leaking.	Actuating valve seat dirty or defective.	Briefly operate actuating valve. Exchange if valve still leaks.
Check valve with limited return flow		
Valve does not hold back closing pressure.	Diaphragm or check valve leaking.	Check diaphragm or valve, replacing if necessary.
Double pressure gauge		
Both pointers do not agree when cut-off pressure reached, out of tolerance ± 0.2 bar.	Wear in transmission mechanism.	Exchange double pressure gauge.
Air escapes at double pressure gauge.	Solder point leaking.	Exchange double pressure gauge.
Warning light		
Light comes on while travelling.	Tank pressure inadequate or non-existent.	Watch double brake pressure gauge until tank pressure is reached. Check compressed air system for leaks, sealing if necessary.
CAUTION: Stop immediately		
	Leakage in hydraulic brake circuit.	Rectify leakage, bleeding brake system if necessary. Check brake reservoir, correcting level if necessary.

Fault	Cause	Remedy
	Brake pad/lining wear	Exchange brake pads/linings. Check brake reservoir, correcting level if necessary.
Warning light does not come on when pressure drops below warning level.	Bulb defective.	Exchange bulb.
	Fault in electrical system.	Rectify electrical fault.
Dual-circuit service brake valve		
In release position, compressed air escapes at vent.	Intake valve leaking.	Exchange intake valve.
In partial and full braking positions, compressed air escapes at vent.	Exhaust valve or sealing rings leaking.	Exchange exhaust valve or sealing rings, exchanging brake valve if necessary.
In lower partial braking position, air escapes from vent.	Sealing ring in actuating cylinder of piston circuit 1 leaking.	Exchange actuating cylinder.
After braking, wheel brakes are slow to release or residual pressure remains in brake system.	No pedal play.	Adjust pedal play.
	Inner stiffness between piston and cylinder bore as well as at fulcrum point of angle lever due to defective bellows, compression weak.	Check brake valve, exchanging if necessary.
Brake graduation not fine enough.	Pedal system, cable connection stiff.	Release pedal system and cable connection.
	Inner stiffness between cylinder and piston bore.	Check brake valve, exchanging if necessary.
Parking brake valve		
At release position, compressed air escapes at vent or bellows.	Exhaust valve or sealing rings leaking.	Exchange exhaust valve and sealing rings.
At partial and full braking positions, compressed air escapes at vent.	Intake valve or sealing ring leaking.	Clean or exchange intake valve. Exchange sealing ring.
Valve does not give satisfactory graduation.	Vent dirty.	Clean vent.
	Inner stiffness at piston.	Overhaul valve, exchanging if necessary.
	Valve stiff at fulcrum point of hand lever.	Release hand lever, exchanging valve if necessary.
	Bellows defective.	Exchange bellows.
In release position, handbrake valve does not provide full air admission.	Poor valve adjustment.	Adjust handbrake valve on test bench.

Fault	Cause	Remedy
Trailer control valve, single-line		
With brakes released, compressed air escapes at vent and at control port.	Exhaust valve leaking, sealing rings leaking.	Clean or exchange valve and valve seat. Exchange sealing ring.
At closed position, compressed air escapes at vent.	Exhaust valve leaking, diaphragm defective.	Clean or exchange valve and valve seat. Exchange diaphragm.
Valve does not provide satisfactory graduation.	Graduating piston sticking. Sealing ring leaking. Snap ring defective.	Check piston and sealing ring, exchanging if necessary. Exchange snap ring.
Assembly does not provide correct advance.	Graduating piston sticking. Spring broken. Diaphragm defective.	Check piston and sealing ring. Exchange spring. Exchange diaphragm.
In release position, pressure in trailer control line rises above 5.4 bar.	Pressure reducing piston or spring defective, intake valve may be leaking.	Check piston and sealing ring, exchanging if necessary. Exchange spring. Check valve and valve seat, exchanging if necessary.
In braking position, pressure rises in trailer control line.	Intake valve leaking.	Check valve and valve seat, exchanging if necessary.
Trailer control line, dual-line		
With service and parking brake released, compressed air escapes at vent.	Intake valve or sealing rings leaking.	Clean or exchange intake valve, exchange sealing rings.
With service and parking brake released, compressed air escapes at vent of service brake valve.	Diaphragm piston leaking to control connection 42.	Exchange control valve.
Response stage of valve too great or graduation not proper. Residual pressure remains in trailer brake line.	Inner stiffness in valve (pistons sticking).	Checking control valve, exchanging if necessary.
In full and partial braking position, compressed air escapes at vent.	Exhaust valve or sealing rings leaking.	Clean or exchange exhaust valve. Exchange sealing rings.
Trailer control valve, dual-line		
With service and parking brake released, residual pressure remains in trailer control line.	Service brake valve does not fully exhaust.	Check pedal play and adjust. Check service brake valve, exchanging if necessary.
	Cable connection stiff.	Release cable connection.
	Handbrake valve does not fully admit air.	Check handbrake valve, exchanging if necessary.
With parking brake applied, compressed air escapes from vent.	Exhaust valve or sealing ring leaking.	Clean or exchange exhaust valve. Exchange sealing rings.

Fault	Cause	Remedy
Piston cylinder (actuating cylinder)		
In braking position, compressed air escapes at vent.	Piston sealing rings leaking.	Exchange sealing rings.
	Cylinder liner scored.	Exchange cylinders.
In partial braking position, compressed air escapes at vent of service brake valve.	Sealing ring at piston circuit 1 of actuating cylinder is leaking.	Exchange sealing ring, replacing cylinder if necessary.
Following braking, piston does not return to released position.	Piston sticking, compression spring defective.	Exchange cylinder.
Spring loaded cylinder		
With parking brake released, compressed air escapes at vent of spring loaded cylinder.	Sealing ring leaking.	Exchange sealing ring or exchange spring loaded cylinder.
Stroke of spring loaded cylinder excessive.	Air gap between brake pad and disk excessive.	Adjust air gap.
Spring loaded cylinder does not return fully to released position.	Release pressure inadequate.	Check parking brake, exchanging if necessary.
Two-way valve		
Compressed air escaping at non-modulated pressure connection line.	Piston slide valve leaking.	Exchange piston valve.
Pressure reducing valve		
Compressed air escapes at vent.	Sealing ring leaking.	Renew sealing ring or exchange valve.
Closing pressure of valve excessive or inadequate.	Spring bias reduced or sealing ring compressed.	Set closing pressure at adjusting screw.
Valve no longer closes. Over-pressure valve opens.	Spring or sealing ring defective. Piston sticking.	Exchange pressure reducing valve.
Coupling head, single-line		
Compressed air escapes in coupled position.	Sealing surfaces defective.	Exchange sealing ring.
Compressed air escapes at coupling head in uncoupled position.	Valve leaking.	Check valves and valve seats or exchange.
Insufficient compressed air to trailer.	Pin of counter coupling head does not press valve fully open.	Renew coupling head with pin.
	Filter clogged.	Clean filter.

Fault	Cause	Remedy
Compressed air escapes at bore of valve of coupling head (special version) in uncoupled position.	Valve body leaking.	Clean valve or exchange coupling head.
Coupling heads, dual-line		
Following coupling, compressed air escapes at sealing surfaces during full braking action.	Sealing surfaces defective.	Exchange sealing rings.
Coupling heads do not lock properly.	Tab washer worn.	Exchange tab washers at both coupling heads.
Following uncoupling, air escapes at supply head.	Valve leaking.	Clean or exchange valve.
	Filter clogged.	Clean filter.
2/2-way valve (breadaway valve)		
Compressed air escapes at aspiration port.	Valve leaking.	Exchange valve.
Pressure drop in supply line during full braking in 2 seconds to 1.5 bar not achieved.	Restriction in valve too big or piston sticking.	Check or exchange valve.
3/2-way valve (actuating valve)		
In release position, compressed air escapes at vent.	Valve leaking.	Clean or exchange valve.
In actuating position, compressed air escapes continuously at vent.	Valve or sealing rings defective.	Clean valve, exchanging if necessary.
Actuating tappit does not fully return to released position.	Spring defective, tappit sticking.	Check or exchange valve.
3/2-way valve (test valve)		
With handbrake in released position, compressed air escapes at rotary knob.	Sealing ring on top of tappit leaking.	Exchange sealing ring or valve.
Handbrake exhausted. Compressed air escapes continuously at vent of handbrake valve.	Sealing ring on bottom of tappit leaking.	Exchange sealing ring or valve.
Handbrake exhausted. Air admitted to test valve. Compressed air escapes continuously at vent of handbrake valve.	Sealing ring in centre of tappit leaking.	Exchange sealing ring or valve.

Fault	Cause	Remedy
4/2-way valve (planetary gear)		
With slow or fast range selected, compressed air escapes at vent.	Sealing rings leaking.	Exchange sealing rings or valve.
	Piston cylinder on transmission leaking.	Exchange piston cylinder or sealing ring.
Actuating tappit does not return fully to released position.	Spring defective, tappit sticking.	Check or exchange valve.
Vehicle is parked in the evening with fast range selected. On the following morning the driver starts the vehicle, shifts into the slow range and drives off. If the ancillary consumers have not yet been supplied with air via the 4-circuit protection valve, the driver is travelling in the fast range although the 4/2-way valve has shifted to the slow range. As soon as the ancillary consumers are supplied with air, the planetary gear shifts four gears into the slow range which may result in consequential damage.	Pressure drop overnight in compressed air supply.	Do not move off until vehicle is ready for the road and ancillary consumers have supply pressure.
4/2-way valve (planetary gear)		
Vehicle is parked in the evening with the fast range selected. The next morning the driver starts the engine and selects the slow range. As a result, sufficient air was in the supply line to the 4/2-way valve to enable the fast range to be disengaged but not the slow range to be engaged. The driver selects a gear at the main transmission with the engine running, the supply reservoirs are filled and he leaves the cab. He has not noticed that the interruption of the power flow to the wheels was only at the planetary range which was not engaged. If the ancillary consumers now get supply pressure, the slow range is correctly selected and the vehicle moves off by itself.	Pressure drop overnight in compressed air supply.	After the engine is running and the compressed air system filled, do not select any gear in the main transmission.
CAUTION: Risk of accident !		

Fault	Cause	Remedy
4/3-way valve (relay valve)		
In released position, compressed air escapes at vent.	Sealing ring at bottom leaking.	Replace sealing ring or exchange valve.
With 4-wheel drive or differential locks engaged, compressed air escapes at vent.	Sealing ring in centre or top leaking.	Replace sealing ring or exchange valve.
Valves does not return to release position.	Spring broken, tappit sticking.	Check or replace valve.

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acc. to DIN 24 300, and with unit connection designations acc. to DIN 74 254

– Provisional Edition –

Unit connection designations acc. to DIN 74 254 (Draft)

Fields of application

This standard applies to air brakes on motor vehicles (including systems with hydraulic power transmission) and their trailers.

The designations should be provided both on the actual units adjacent to the connections and in the brake diagrams adjacent to the power connections indicated there.

Designations

The designations consist of one or two digits. The meanings of the **first digit** are:

- 0 Intake connection
- 1 Power inflow
- 2 Power outflow (not for outflow to atmosphere)
- 3 Connection to atmosphere (for bleeding)
- 4 Control connection (input into unit)

A second digit must be used whenever several similar connections, e.g. for several circuits, are possible or provided. The **second digit** ought logically to begin at 1, e. g. 21, 22, 23 etc. Several similar connections from one chamber are to be given one and the same designation.

If a connection is able to perform several functions, two (first) digits will be required, separated by a horizontal line.

Example

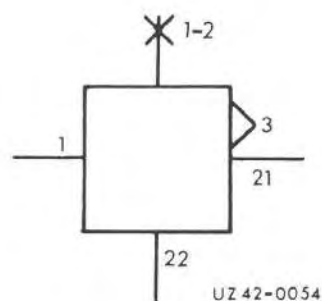
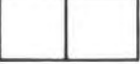
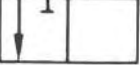
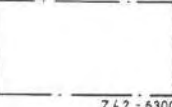


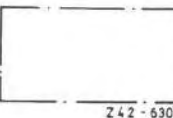
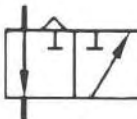
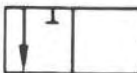
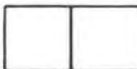
Fig. 1: Pressure regulator with filling and switching connections

- 1 Power inflow from compressor
- 1 – 2 Power inflow for a valve used to charge air brake system, or optionally power outflow for a valve used to inflate tires
- 3 Connection to atmosphere (for bleeding)
- 21 Power outflow to accumulator (air tank)
- 22 Power outflow (switching connection)

Designation	Symbol
Pressure source	
Power line	
Control line	
Hose	
Electric line	
Line connection	
Line crossing	
Lever, rod, shaft	
Spring	
Muscular actuation	
a) general	
b) Button	
c) Lever	
d) Pedal	
Mechanical actuation	
a) Spring	
b) Button	

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Designation	Symbol
Electrical actuation	
Pneumatic actuation	
a) Pressure rise	
b) Pressure drop	
Detent	
Pressure medium designation	
a) pneumatic	
b) hydraulic	
Direction of flow	
Valves with several fixed service positions (directional control valves)	
e. g. 2/2-way valve, 3/2-way valve, etc.	
The number of squares corresponds to the number of service positions.	
Within the squares, the lines indicate the actual power lines, and the arrows the directions of flow. Shutoffs are identified by means of crossing lines within the squares.	
All other service positions are obtained by shifting the squares until the power lines are aligned with the connections of the next square.	
A square surrounded by a fine dash-and-dot line indicates several parts forming a block or design group.	



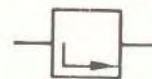
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Valves without fixed service positions

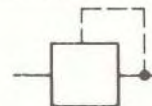
Valves which are required to assume a number of intermediate positions between two end functions are represented by a plain square.



If, during a change in position, the inflow or outflow continues to communicate with a particular connection, the end of the arrow will bear a crossing line which will be left attached to the arrow when the rectangular figure is shifted. The change in position is to be depicted in the same manner as for valves with several fixed positions, the rectangular figure being shifted with the lines and arrow at right-angles to the connections.

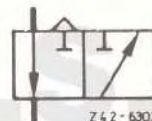


Broken line = internal control path
(not an air line)



3/2-way valve means:

- 3 controlled ports
- 2 service positions



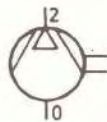
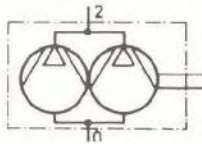
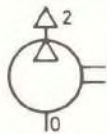
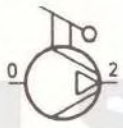
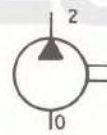
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Survey

No.	Designation
1	Compressor, pumps and vacuum pumps
2	Pressure regulators
3	Defrosters
4	Protection valves
5	Power accumulators
6	Water separators and driers
7	Overflow valves
8	Check and shutoff valves
9	Warning lights
10	Pressure switches, electric
11	Pressure gauges
12	Warning devices, acoustic
13	Service brake valves
14	Parking brake valves
15	Brake valves, hand operated
16	Relay valves
17	Trailer control valves for single-line brake systems
18	Trailer control valves for dual-line brake systems
19	Single-acting cylinders (hydraulic)
20	Single-acting cylinders (pneumatic)
21	Double-acting cylinders (pneumatic and hydraulic)
22	Cylinders, combined (pneumatic)
23	Brake boosters
24	Pressure ratio valves
25	Pressure control valves
26	Brake pressure modulators, automatic
27	Brake servo units
28	Non-return valves
29	Pressure reducing valves
30	Pressure limiting valves
31	Electrical unit
32	Directional control valve with 3 positions
33	Directional control valves with 2 positions
34	Reservoir (hydraulic)
35	Coupling heads
36	Filters
37	Throttle valves
38	Pressure connection points
39	Bleed point
40	One-way restrictors
41	Directional control valve with 3 positions
42	Directional control valves with 4 positions
43	Air spring bellows
44	(not yet specified)
45	Directional control valves with 2 positions
46	Linkages

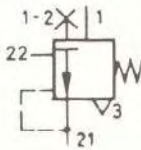
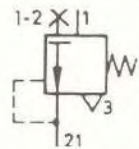
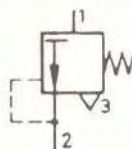
Id.	Designation	Symbol
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Compressor, pumps and vacuum pumps

1.1	Single-cylinder compressor	
1.2	Two-cylinder compressor	
1.3	Vacuum pump	
1.4	Compressor, hand-operated	
1.5	Hydraulic pump (with 1-way delivery)	

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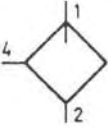
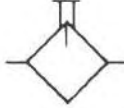
2 Pressure regulators

2.1	— With combined charging/discharging connection and with switching connection	
2.2	— With combined charging/discharging connection	
2.3	— Without auxiliary connections	

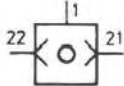
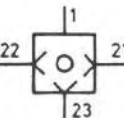
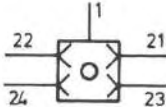
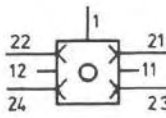
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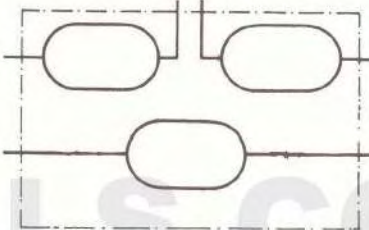
No.	Designation	Symbol
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3 Defrosters

3.1	— With wick	
3.2	— Automatic with control connection	
3.3	— Manually actuated	 UZ 43-0151

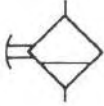
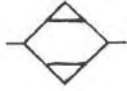
4 Protection valves

4.1	Two-circuit	
4.2	Three-circuit	
4.3	Four-circuit — with 5 connections	
4.4	Four-circuit — with 7 connections	 UZ 43-0152

No.	Designation	Symbol
5 Power accumulators		
5.1	Single-chamber air accumulator (DIN 74 281)	
5.2	Two-chamber air accumulator (DIN 74 281)	
5.3	Three-chamber air accumulator (DIN 74 281)	 Containers may be arranged as required
5.4	Single-chamber vacuum accumulator	
5.5	Hydraulic accumulator	 UZ 43-0153

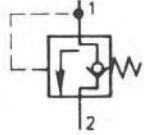
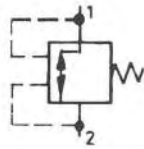
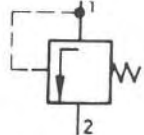
No.	Designation	Symbol
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6 Water separators and driers

6.1	Water separator, manually actuated	
6.2	Water separator with automatic drainage and control connection	
6.3	Water separator with automatic drainage	
6.4	Drier	

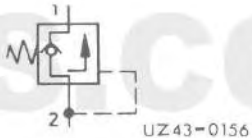
UZ43-0154

7 Overflow valves

7.1	With return flow (DIN 74 279)	
7.2	With limited return flow (DIN 74 279)	
7.3	Without return flow (DIN 74 279)	

No.	Designation	Symbol
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8 Check and shutoff valves

8.1	Check valve	
8.2	Check valve with counter-pressure spring	
8.3	Shutoff valve (DIN 74 293)	
8.4	Check valve with limited return flow	
8.5	Prepressure valve	

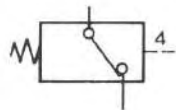
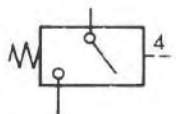
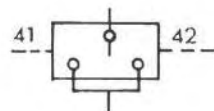
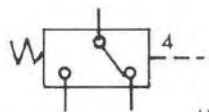
9 Warning lights (DIN 40 708)

9.1	– red	
9.2	– white	
9.3	– Stoplight	
9.4	– green	
9.5	– yellow	
9.6	– solo	
9.7	– Trailer	
9.8	– Information according to operation	

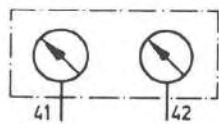
UZ43 -0157

No.	Designation	Symbol
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10 Pressure switches, electric

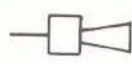
10.1	Cut-out switch	
10.2	Cut-in switch	
10.3	Differential pressure switch	
10.4	Changeover	 UZ 43 - 0158

11 Pressure gauges

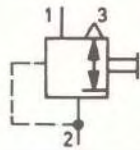
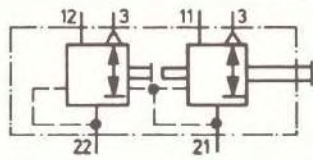
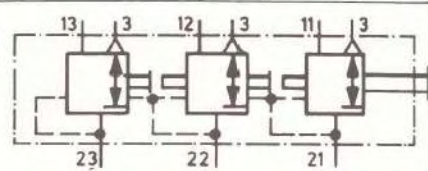
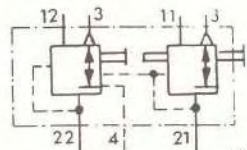
11.1	Single	
11.2	Double	 UZ 43 - 0159

No.	Designation	Symbol
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12 Warning devices, acoustic

12.1	Buzzer (DIN 40 708)	
12.2	Horn (DIN 40 717)	 UZ43-0160
12.3	Rod sensor	 UZ43-0228

13 Service brake valves

13.1	Single-circuit	
13.2	Dual-circuit	
13.3	Three-circuit	
13.4	Dual-circuit – with ALB modulation	 UZ43-0161

No.	Designation	Symbol
14 Parking brake valves		
14.1	For spring-loaded cylinder (step type)	
14.2	For spring-loaded cylinder (step type) and trailer control valve (ECC)	
14.3	For spring-loaded cylinder (step type) and "halt" brake	
14.4	For spring-loaded cylinder (step type) with pressure limitation at output	
14.5	For spring-loaded cylinder, not step type Step type auxiliary brake	
14.6	For spring-loaded cylinder with linkage control	
14.7	For spring-loaded cylinder (step type) and trailer control (UK)	
14.8	For spring-loaded cylinder (step type) and auxiliary valve	

No.	Designation	Symbol
14.9	For spring-loaded cylinder (step type) as well as trailer control and auxiliary valve	

15 Brake valves, hand operated

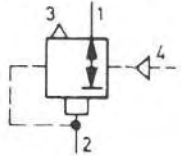
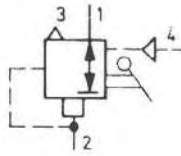
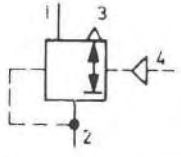
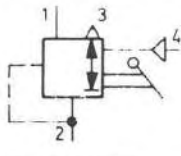
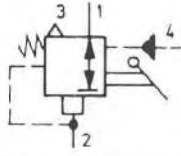
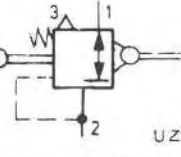
15.1	Anti-jackknife brake valve	
15.2	"Halt" brake valve with pressure limitation at output	

16 Relay valves

16.1	Admitting	
16.2	Exhausting (control valve for France)	
16.3	Admitting and with additional lever actuation	
16.4	Admitting and with hydraulic control	

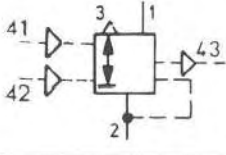
No.	Designation	Symbol
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17 Trailer control valves for single-line brake system

17.1	Air-controlled with pressure limitation at output	
17.2	Air-controlled with pressure limitation at output and additional lever actuation	
17.3	Air-controlled, without pressure limitation at output	
17.4	Air-controlled, without pressure limitation at output, with additional lever actuation	
17.5	Hydraulic control, with pressure limitation at output and additional lever actuation	
17.6	Servo-controlled, without pressure limitation at output, mechanically actuated	

UZ 43-0165

18 Trailer control valves for dual-line brake systems

18.1	Three-circuit control (admitting and exhausting)	
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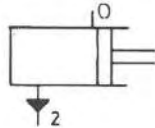
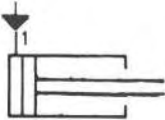
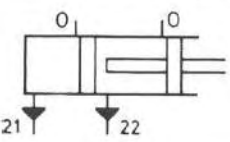
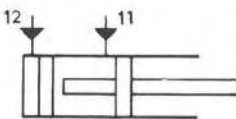
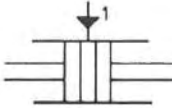
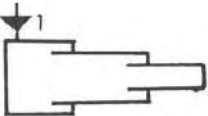
UZ 43-0166

No.	Designation	Symbol
18.2	Two-circuit control (admitting)	
18.3	Two-circuit control (admitting and exhausting)	
18.4	Three-circuit control for trucks (ECC) (with breakaway protection)	
18.5	Three-circuit control for tractive unit (ECC) (with breakaway protection)	

UZ 43 - 0166/1

No	Designation	Symbol
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19 Single-acting cylinders (hydraulic)

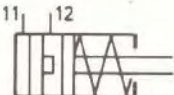
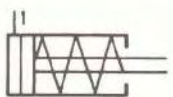
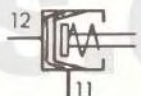
19.1	Master cylinder, single-circuit	
19.2	Slave cylinder, single-circuit	
19.3	Master cylinder, dual-circuit	
19.4	Slave cylinder, dual-circuit	
19.5	Wheel cylinder	
19.6	Telescopic cylinder	

UZ 43 - 0167

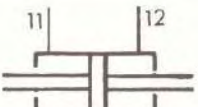
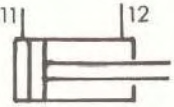
20 Single-acting cylinders (pneumatic)

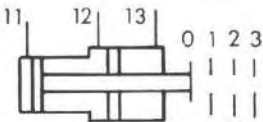
20.1	Single-circuit piston cylinder	
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UZ 43 - 0168

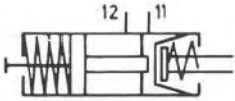
No.	Designation	Symbol
20.2	Single-circuit diaphragm cylinder	
20.3	Dual-circuit piston cylinder	
20.4	Spring-loaded cylinder, pushing	
20.5	Spring-loaded cylinder, pulling	
20.6	Dual-circuit diaphragm cylinder	
20.7	Telescopic cylinder	 UZ 43-0168/1

21 Double-acting cylinders (pneumatic and hydraulic)

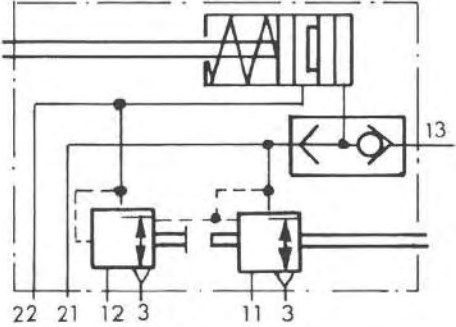
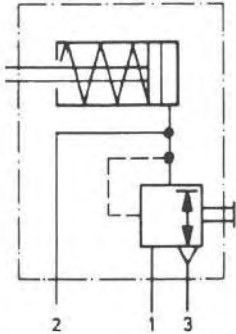
21.1	— With double-ended piston rod	
21.2	— With single-ended piston rod	 UZ 43-0169

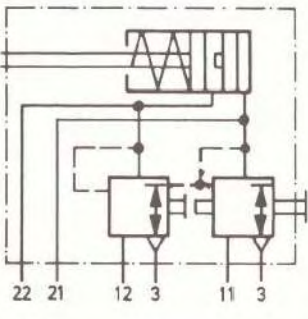
No.	Designation	Symbol
21.3	Four-position cylinder	 UZ43-0169/1

22 Double brake cylinder (pneumatic)

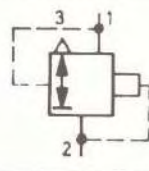
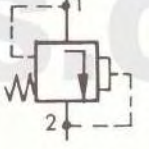
22.1	Diaphragm spring-loaded cylinder with mechanical release	 UZ43-0192
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23 Brake boosters (power-assisted)

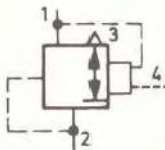
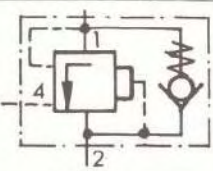
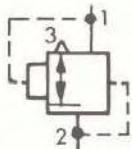
23.1	— Dual-circuit with two-way valve	 UZ43-0170
23.2	— Single-circuit	 UZ43-0170

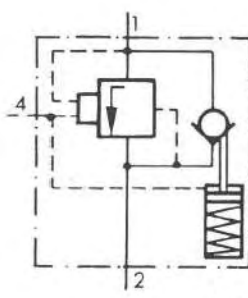
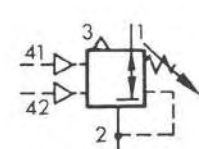
No.	Designation	Symbol
23.3	— Dual-circuit	

24 Pressure ratio valves

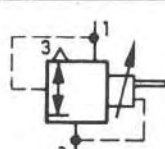
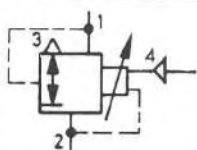
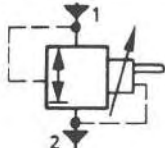
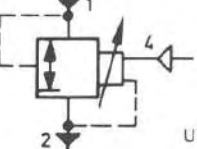
24.1	— With special characteristic and exhausting	
24.2	— With special characteristic without exhausting	

25 Pressure control valves

25.1	— Pneumatic	
25.2	— With return flow	
25.3	Reducing valve	

No.	Designation	Symbol
25.4	– Hydraulic with throughcontrol	 UZ43-0229
25.5	– Pneumatic with 2 control connections	 UZ43-0230

26 Brake pressure modulators, automatic

26.1	– Mechanically actuated for air brakes	
26.2	– Pneumatically actuated for air brakes	
26.3	– Mechanically actuated for hydraulic brakes	
26.4	– Pneumatically actuated for hydraulic brakes	 UZ43-0246

No.	Designation	Symbol
26.5	— Mechanically actuated with hydraulic control connection for output control	
26.6	— Pneumatically actuated for continuous brake with 2 control connections (average pressure is modulated)	
26.7	— With relay valve	
26.8	— Mechanically actuated for hydraulic brakes (tandem version)	
26.9	— Pneumatically actuated for compressed air brake with 3 control connections	

No.	Designation	Symbol
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27 Brake servo units

27.1	— Single-circuit (for auxiliary brake, full pedal force applied)	
27.2	— Dual-circuit (for auxiliary brake, partial pedal force applied)	
27.3	— Single-circuit vacuum (for auxiliary brake, full pedal force applied)	
27.4	— Single-circuit vacuum with auxiliary cylinder (for auxiliary brake, full pedal force applied)	

UZ43-0173

No.	Designation	Symbol
27.5	— Single-circuit (for auxiliary brake, partial pedal force applied)	
27.6	Handbrake servo unit (for parking brake)	
27.7	— Single-circuit with auxiliary cylinder capable of sequencing (for auxiliary brake, full pedal force applied)	

UZ 43-0173/1

28 Non-return valves

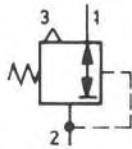
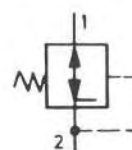
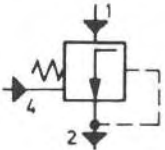
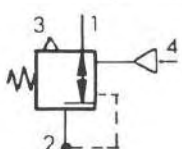
28.1	Shuttle valve	
28.2	Rapid-exhaust valve	
28.3	Shuttle valve, pretensioned	

UZ 43-0174

UZ 43-0232

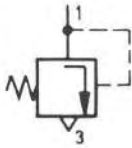
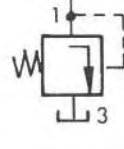
No.	Designation	Symbol
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29 Pressure reducing valves

29.1	— With exhausting	
29.2	— Without exhausting (separator)	
29.3	— With control connection, without outlet opening	
29.4	— With control connection and exhausting	

UZ 43-0175

30 Pressure limiting valves

30.1	Safety valve	
30.2	Pressure limiting valve	

UZ 43-0176